# Checkpoint 6 Maths — Practice Test 2 — Mark Scheme

Paper 1: 45 minutes · no calculator · Paper 2: 45 minutes · calculator allowed · Total 80 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

## Paper 1 — 45 minutes — 40 marks

**1** **[1]**

What is the value of the digit **7** in the number **3.472**?

- ☐ A) 7 tenths
- ☐ B) 7 hundredths
- ☐ C) 7 thousandths
- ☐ D) 7 ones

**Answer:** B) 7 hundredths
**Notes / working:** 3.472 = 3 ones + 4 tenths + 7 hundredths + 2 thousandths. The 7 is in the **second** place after the decimal point, so it is worth **7 hundredths** (0.07). Common mistake: counting the 4 as the hundredths place — the first place after the point is tenths.

**2** The freezer in a supermarket is at −18 °C. The fridge next to it is 22 degrees warmer. **[1]**

What is the temperature in the fridge?

**Answer:** 4 °C
**Notes / working:** Start at −18 and count on 22. From −18 to 0 is 18 degrees; there are 22 − 18 = 4 degrees left, so the fridge is at **4 °C**. Common mistake: −18 − 22 = −40 (going the wrong way — warmer means count **up**).

**3** The numbers in this sequence go down by the same amount each time. **[1]**
1.4, 1.0, 0.6, 0.2, …

Write the next number in the sequence.

**Answer:** -0.2
**Notes / working:** Each step is −0.4 (1.4 → 1.0 → 0.6 → 0.2). The next number is 0.2 − 0.4 = **−0.2**. Count back past zero: 0.2 − 0.2 = 0, then 0.2 more takes you to −0.2. Do not accept 0 (stopping at zero) or 0.2.

**4** **[1]**

Write the missing number.
4.6 × ☐ = 4600

> **Answer:** 1000
> **Notes / working:** 4.6 → 4600: the digit 4 moves from the ones place to the thousands place, which is **3 places** to the left. Moving digits 3 places to the left means multiplying by **1000**. Check: 4.6 × 1000 = 4600 ✓.

**5** **[1]**

Write the missing number.
5.208 = 5 + 0.2 + ☐

> **Answer:** 0.008
> **Notes / working:** 5.208 = 5 ones + 2 tenths + 0 hundredths + 8 thousandths = 5 + 0.2 + **0.008**. Common mistake: writing 0.08 — the 8 is in the third decimal place, so it is 8 thousandths.

**6** **[1]**

Work out.
36 − 4 × 5 + 2

> **Answer:** 18
> **Notes / working:** Multiplication comes before addition and subtraction: 4 × 5 = 20. Then work left to right: 36 − 20 = 16, 16 + 2 = **18**. Common mistake: working left to right from the start, (36 − 4) × 5 + 2 = 162.

**7** Here is a calculation. It is not correct yet. **[1]**
6 + 4 × 5 − 2 = 30

Write brackets in the calculation to make it correct. Write the whole calculation on paper.

> **Answer:** 1 mark: (6 + 4) × (5 − 2) = 30. Both pairs of brackets are needed. Do not accept any other placing of brackets.
> **Notes / working:** Try the options. (6 + 4) × 5 − 2 = 50 − 2 = 48 ×. 6 + 4 × (5 − 2) = 6 + 12 = 18 ×. **(6 + 4) × (5 − 2)** = 10 × 3 = 30 ✓. Brackets are worked out first, so they change the order of operations.

**8** A cinema sells 1375 tickets every week. **[2]**

How many tickets does it sell in 24 weeks? Show your working.

**Answer:** 2 marks for 33 000 (accept 33000). If 0 marks: M1 for a complete correct method, e.g. 1375 × 20 + 1375 × 4, a grid method or long multiplication, with no more than one arithmetic error.
**Notes / working:** 1375 × 24 = 1375 × 20 + 1375 × 4 = 27 500 + 5500 = **33 000** tickets. Estimate to check: 1400 × 24 ≈ 1400 × 25 = 35 000, so 33 000 is sensible. (M1 for the method, A1 for 33 000.)

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**9** 456 people are travelling to a festival by bus. Each bus can carry 32 people. **[2]**

How many buses are needed? Show your working.

**Answer:** 2 marks for 15. If 0 marks: M1 for 456 ÷ 32 attempted with 14 r 8 or 14.25 seen (or 32 × 14 = 448 seen). Do not accept 14 for the final answer.
**Notes / working:** 456 ÷ 32: 32 × 10 = 320, 456 − 320 = 136; 32 × 4 = 128, 136 − 128 = 8. So 456 ÷ 32 = 14 remainder 8. Fourteen buses carry 448 people and 8 people are left, so one more bus is needed: **15 buses**. Common mistake: answering 14 (ignoring the remainder) or 14.25 (you cannot have part of a bus).

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**10** Bus route 12 leaves the bus station every 6 minutes. Bus route 30 leaves every 8 minutes. Both buses leave together at 07:00. **[1]**

At what time do both buses next leave together?

**Answer:** 07:24 or 7:24 or 0724 or 07.24 or 7.24 or 7:24am or 07:24am or 7.24am
**Notes / working:** Route 12 leaves at 6, 12, 18, **24**, 30 … minutes past 7. Route 30 leaves at 8, 16, **24**, 32 … minutes past 7. The lowest common multiple of 6 and 8 is 24, so they next leave together at **07:24**. Common mistake: 6 × 8 = 48 gives a common multiple, but not the first one.

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**11** Omar says, "2016 is divisible by 9." **[1]**

Is Omar correct? Explain how you know without doing the division.

**Answer:** 1 mark: Yes, with a correct reason, e.g. the digit sum 2 + 0 + 1 + 6 = 9, and 9 is divisible by 9. Do not accept doing the division (2016 ÷ 9 = 224), because the question asks for a reason without dividing. Do not accept 'Yes' with no reason or with a wrong reason (e.g. 'it ends in 6').
**Notes / working:** Test of divisibility by 9: add the digits. 2 + 0 + 1 + 6 = **9**, which is a multiple of 9, so 2016 **is** divisible by 9. Omar is correct. (Check: 9 × 224 = 2016.)

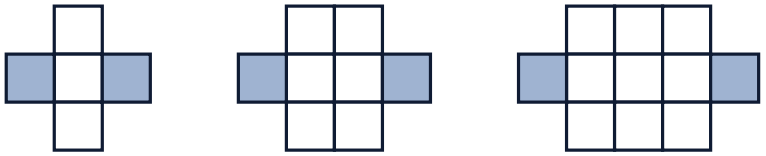
**12** **[1]**

Which number is both a **square number** and a **cube number**?

- ☐ A) 8
- ☐ B) 16
- ☐ C) 27
- ☐ D) 64

**Answer:** D) 64
**Notes / working:** 64 = 8 × 8 (square) and 64 = 4 × 4 × 4 (cube). 8 and 27 are cube numbers only ($2^3$ and $3^3$); 16 is a square number only ($4^2$). The answer is **64**.

**13** Laila makes patterns with white tiles and grey tiles. Each new pattern has 3 more white tiles than the one before. **[1]**



Pattern 1 Pattern 2 Pattern 3

(a) How many tiles (white and grey) are in **Pattern 10**?

**Answer:** 32
**Notes / working:** Count the tiles: Pattern 1 has 5, Pattern 2 has 8, Pattern 3 has 11 — 3 more each time. Every pattern has 3 white tiles for each position number and 2 grey tiles, so Pattern 10 has 3 × 10 + 2 = 30 + 2 = **32** tiles.

(b) Which rule gives the number of tiles in **any** pattern? **[1]**

- ☐ A) number of tiles = 3 × position + 2
- ☐ B) number of tiles = position + 3
- ☐ C) number of tiles = 2 × position + 3
- ☐ D) number of tiles = 5 × position

**Answer:** A) number of tiles = 3 × position + 2
**Notes / working:** Test each rule with position 1 (5 tiles) and position 2 (8 tiles). 3 × 1 + 2 = 5 ✓ and 3 × 2 + 2 = 8 ✓. The other rules fail: 1 + 3 = 4, 2 × 2 + 3 = 7, 5 × 2 = 10. The answer is **3 × position + 2** — the 3 comes from adding 3 tiles each time (repeated addition = multiplication).

(c) Can one of Laila's patterns have exactly **40** tiles? Explain how you know. **[1]**

> **Answer:** 1 mark: No, with a correct reason, e.g. 40 − 2 = 38 and 38 is not a multiple of 3 / 38 ÷ 3 is not a whole number; or Pattern 12 has 38 tiles and Pattern 13 has 41 tiles. Do not accept 'No' with no reason.
> **Notes / working:** If a pattern had 40 tiles, then 3 × position + 2 = 40, so 3 × position = 38. But 38 is not in the 3 times table (3 × 12 = 36, 3 × 13 = 39). Pattern 12 has 38 tiles and Pattern 13 has 41 tiles, so **no** pattern has exactly 40 tiles.

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**14** The rule for a sequence is: **[1]**
term = position × position + 3

What is the **5th** term of the sequence?

> **Answer:** 28
> **Notes / working:** Position 5: 5 × 5 + 3 = 25 + 3 = **28**. (The first terms are 4, 7, 12, 19, 28 — the square numbers 1, 4, 9, 16, 25 with 3 added.) Common mistake: 5 × 2 + 3 = 13 (doubling instead of squaring).

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**15** At the supermarket, a bag of oranges costs **p** pounds. A bag of apples costs 15 pounds more than a bag of oranges. **[1]**

Write an expression for the cost of a bag of apples, in pounds.

> **Answer:** p+15 or 15+p
> **Notes / working:** '15 pounds more than p' means p add 15, so the cost is **p + 15** pounds. The letter p stands for a number that can vary. Do not write 15p — that means 15 × p.

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**16** **[1]**

Which fraction is 18/24 written in its **simplest form**?

- ☐ A) 9/12
- ☐ B) 6/8
- ☐ C) 3/4
- ☐ D) 2/3

> **Answer:** C) 3/4
> **Notes / working:** The highest common factor of 18 and 24 is 6. 18 ÷ 6 = 3 and 24 ÷ 6 = 4, so 18/24 = **3/4**. 9/12 and 6/8 are equal to 18/24 but can still be simplified; 2/3 is not equal to 18/24.

**17** **[1]**

Write **17 ÷ 5** as a mixed number.

**Answer:** 3 2/5 or 3and2/5
**Notes / working:** 17 ÷ 5 = 3 remainder 2. The remainder 2 is shared between 5, giving 2/5. So 17 ÷ 5 = 17/5 = **3 2/5**.

**18** **[1]**

Work out 3/5 of 45.

**Answer:** 27
**Notes / working:** Find 1/5 first: 45 ÷ 5 = 9. Then 3/5 is 3 × 9 = **27**.

**19** Salma has 2 1/4 kg of rice. She uses 5/6 kg of it to make koshari. **[2]**

How much rice does she have left? Give your answer as a mixed number. Show your working.

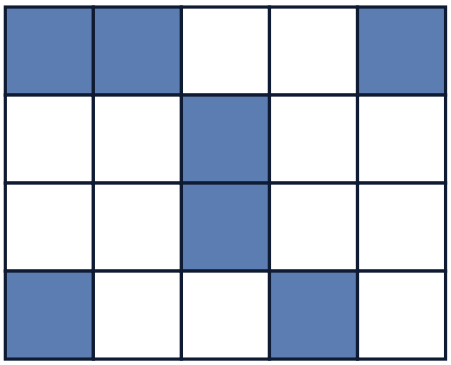
**Answer:** 17/12 kg
**Notes / working:** Use a common denominator of 12: 2 1/4 = 2 3/12 = 27/12 and 5/6 = 10/12. Then 27/12 − 10/12 = 17/12 = **1 5/12 kg**. Estimate: about 2 − 1 = 1, so a bit more than 1 kg is sensible. (M1 for both fractions written with the same denominator, e.g. 27/12 and 10/12; A1 for 1 5/12, accept 17/12.)

**20** **[1]**

Work out 4/5 ÷ 2. Give your answer as a fraction.

**Answer:** 2/5 or 4/10
**Notes / working:** Dividing by 2 means finding half. Half of 4 fifths is **2 fifths**: 4/5 ÷ 2 = 2/5. (Or 4/5 ÷ 2 = 4/10 = 2/5.) Common mistake: 8/5 — that is 4/5 × 2.

**21** Here is a tile design made of equal squares. **[1]**



What **percentage** of the design is shaded?

**Answer:** 35 %
**Notes / working:** There are 5 × 4 = 20 squares and 7 are shaded, so 7/20 is shaded. 7/20 = 35/100 = **35%** (multiply top and bottom by 5).

**22** Hana says, "0.45 is bigger than 0.5 because 45 is bigger than 5." **[1]**

Explain why Hana is **wrong**.

**Answer:** 1 mark for a correct explanation, e.g. 0.5 has 5 tenths but 0.45 has only 4 tenths; or 0.5 = 0.50 and 50 hundredths is more than 45 hundredths; or 0.5 = 50% and 0.45 = 45%. Do not accept '0.5 is bigger' with no reason.
**Notes / working:** Compare the tenths first: 0.5 has **5 tenths**, 0.45 has only **4 tenths**, so 0.5 is bigger. Or write both with two decimal places: 0.50 > 0.45 because 50 hundredths > 45 hundredths. Hana compared the digits as if they were whole numbers.

**23** At the supermarket, Karim buys a bag of rice for E£34.75 and a packet of tea for E£18.50. He pays with a E£100 note. **[2]**

How much change should he get? Show your working.

**Answer:** 46.75 E£
**Notes / working:** Total: 34.75 + 18.50 = 53.25. Change: 100 − 53.25 = **E£46.75**. Check by adding on: 53.25 + 0.75 = 54, 54 + 46 = 100, so the change is 46 + 0.75 = 46.75 ✓. (M1 for 53.25 seen or 100 − 34.75 − 18.50; A1 for 46.75.)

**24** One floor tile costs E£4.35. **[1]**

How much do 6 tiles cost?

> **Answer:** 26.1 E£
> **Notes / working:** 4.35 × 6 = 4 × 6 + 0.35 × 6 = 24 + 2.10 = **E£26.10**. Estimate: 4 × 6 = 24, so about E£26 is sensible.

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**25** **[1]**

Round **7.46** to the nearest tenth.

> **Answer:** 7.5
> **Notes / working:** 7.46 is between 7.4 and 7.5. Look at the hundredths digit: 6 is 5 or more, so round **up** to **7.5**. Common mistake: writing 7.4 or 7.47.

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**26** A film at the cinema lasts 1 hour 45 minutes. **[1]**

Write 1 hour 45 minutes as a decimal number of hours.

> **Answer:** 1.75 hours
> **Notes / working:** 45 minutes = 45/60 of an hour = 3/4 hour = 0.75 hours. So 1 hour 45 minutes = **1.75 hours**. Common mistake: 1.45 hours — there are 60 minutes in an hour, not 100.

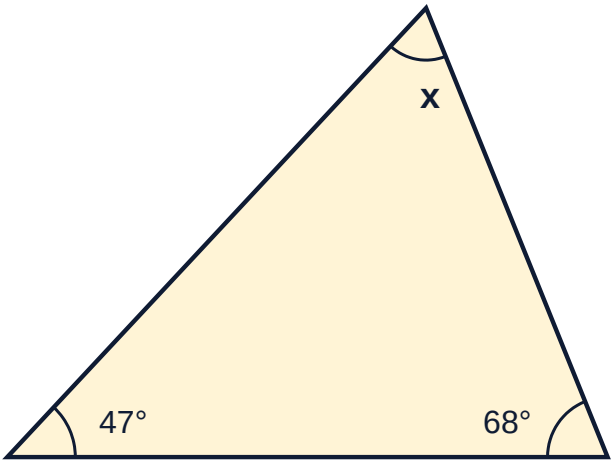
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**27** **[1]**

Sketch a quadrilateral that has **exactly one pair of parallel sides** and **exactly two right angles**. Draw it on paper.

> **Answer:** 1 mark: a right-angled trapezium — one pair of parallel sides of different lengths and one side perpendicular to both of them (two right angles next to each other). Sketch need not be accurate, but must clearly not be a rectangle or square.
> **Notes / working:** The shape is a **right-angled trapezium**: draw two parallel horizontal lines of different lengths, join their left ends with a vertical line (this makes two right angles) and join the right ends with a sloping line. A rectangle has two pairs of parallel sides and four right angles, so it does not fit.

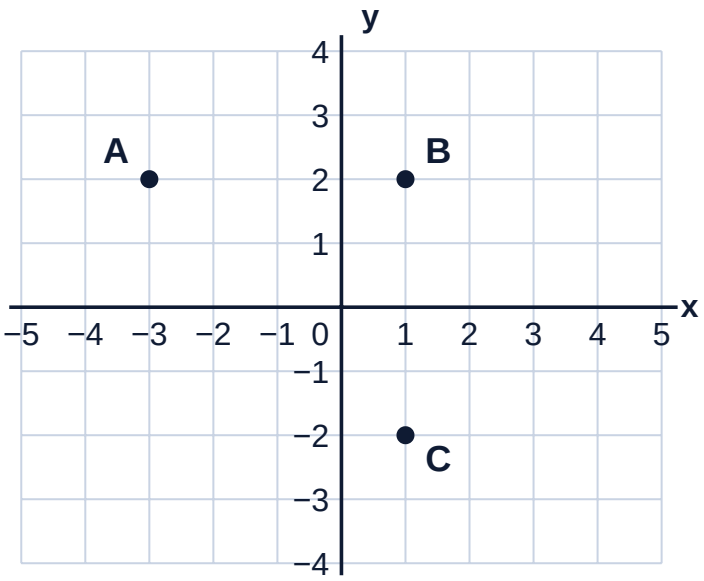
**28** Here is a triangle-shaped tile. **[1]**




Work out the size of angle **x**.

**Answer:** 65 °
**Notes / working:** The angles in a triangle add up to 180°. 47° + 68° = 115°, so x = 180° − 115° = **65°**.

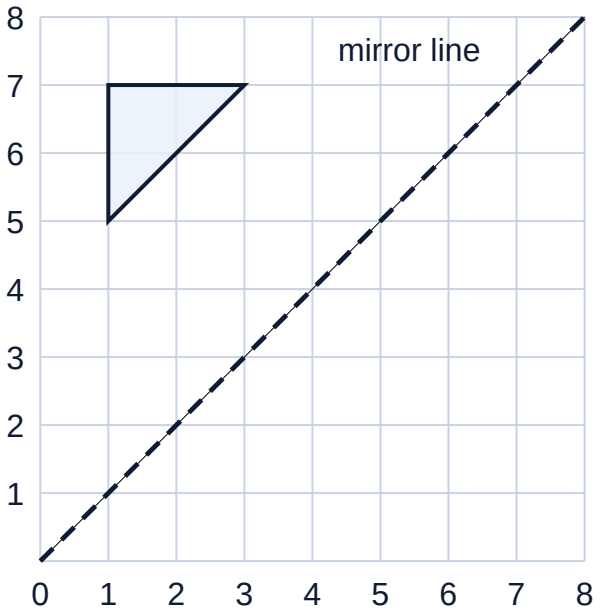
**29** A, B and C are three corners of a square. D is the fourth corner. **[1]**




Write the coordinates of point **D**.

**Answer:** (-3,-2) or (−3,−2) or -3,-2 or −3,−2 or (-3, -2) or (−3, −2)
**Notes / working:** A is (−3, 2), B is (1, 2) and C is (1, −2). AB is 4 squares across and BC is 4 squares down. D must be 4 squares below A (same x as A, same y as C): **D = (−3, −2)**. Common mistake: writing (−2, −3) — x comes first.

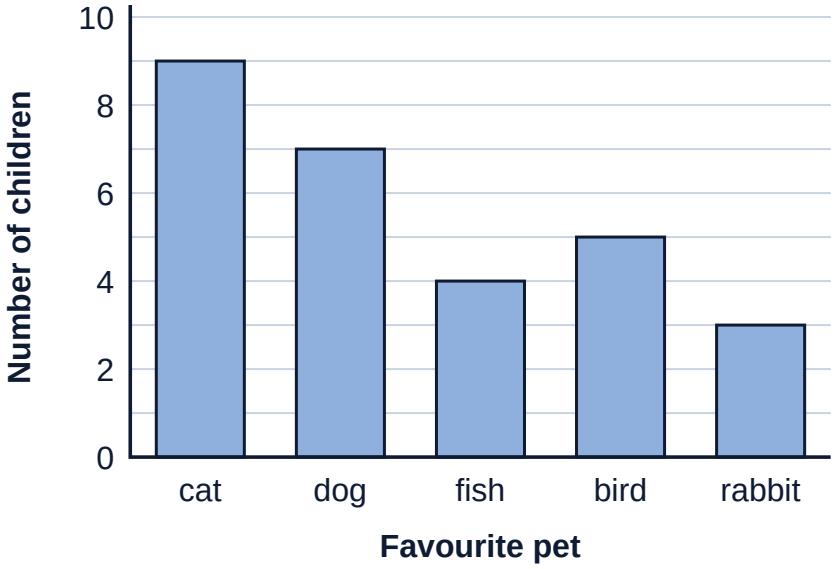
**30** Here is a triangle on a square grid and a diagonal mirror line. **[1]**




Copy the grid and draw the reflection of the triangle in the mirror line.

**Answer:** 1 mark: image triangle with vertices at (5, 1), (7, 1) and (7, 3), using the numbers on the grid edges (across, then up). The right angle of the image is at (7, 1). Do not accept a translation or a reflection in a vertical or horizontal line.
**Notes / working:** Reflect each corner across the diagonal line: each point's 'across' and 'up' numbers swap. (1, 5) → (5, 1); (1, 7) → (7, 1); (3, 7) → (7, 3). Join the three new points. Check: each corner and its image are the same distance from the mirror line, and the line joining them crosses the mirror at a right angle.

**31** Every child in Class 6B said their favourite pet. The bar chart shows the results. **[1]**




(a) How many children are in Class 6B?

**Answer:** 28
**Notes / working:** Read every bar: cat 9, dog 7, fish 4, bird 5, rabbit 3. Total = 9 + 7 + 4 + 5 + 3 = **28** children.

(b) Which favourite pet is the **mode**? **[1]**

**Answer:** cat or cats or a cat
**Notes / working:** The mode is the most common answer — the tallest bar. **Cat** has 9 children, more than any other pet.

(c) Hamza says, "More than half of the class chose a cat or a dog." Is Hamza correct? Show how you know. **[1]**

**Answer:** 1 mark: Yes, with working, e.g. 9 + 7 = 16 and half of 28 is 14 (16 > 14). Accept FT from their total in (a). Do not accept 'Yes' with no numbers.
**Notes / working:** Cat + dog = 9 + 7 = 16 children. Half of 28 is 14. 16 is more than 14, so **Hamza is correct**.

**32** In a game, a bag has 6 cat stickers, 3 dog stickers and 1 fish sticker. Rania takes one sticker without looking. **[1]**

Which word best describes the chance that Rania takes a cat sticker **or** a dog sticker?

- ☐ A) impossible
- ☐ B) unlikely
- ☐ C) even chance
- ☐ D) likely

**Answer:** D) likely
**Notes / working:** 9 of the 10 stickers are cats or dogs, so this will happen 9 times out of 10 — it is **likely** (but not certain, because the fish sticker could be taken).

## Paper 2 — 45 minutes — 40 marks

**1** A bus leaves Saint Catherine early in the morning when the temperature is −3 °C. It arrives in Sharm El-Sheikh, where the temperature is 24 °C. **[1]**

By how many degrees is Sharm El-Sheikh warmer?

**Answer:** 27 °C
**Notes / working:** From −3 to 0 is 3 degrees, and from 0 to 24 is 24 degrees. 3 + 24 = **27 degrees**. (24 − (−3) = 27.) Common mistake: 24 − 3 = 21.

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**2** A supermarket sells 2485 bottles of water each day. **[1]**

How many bottles does it sell in 28 days?

**Answer:** 69580
**Notes / working:** 2485 × 28 = **69 580** bottles. Estimate to check: 2500 × 30 = 75 000, so 69 580 is sensible.

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**3** Nour has 36 red tiles and 48 blue tiles. She puts them into packs. Every pack must be the same, and she must use all the tiles. **[1]**

What is the **greatest** number of packs she can make?

**Answer:** 12
**Notes / working:** The number of packs must be a common factor of 36 and 48. Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. The highest common factor is **12** (each pack has 3 red and 4 blue tiles).

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**4** **[1]**

Which of these numbers is divisible by **6**?

- ☐ A) 334
- ☐ B) 405
- ☐ C) 552
- ☐ D) 718

**Answer:** C) 552
**Notes / working:** A number is divisible by 6 if it is even **and** divisible by 3 (digit sum a multiple of 3). 334: digit sum 10 ×. 405: odd ×. **552**: even, digit sum 12 ✓ (552 ÷ 6 = 92). 718: digit sum 16 ×.

**5** **[1]**

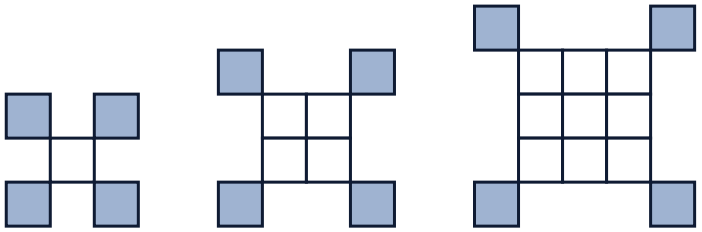
Three of these calculations give the same answer. Which one gives a **different** answer?

- ☐ A) 25 × 4 × 37
- ☐ B) 37 × 25 × 4
- ☐ C) 100 × 37
- ☐ D) 25 + 4 × 37

**Answer:** D) 25 + 4 × 37
**Notes / working:** Multiplication can be done in any order (commutative and associative laws): 25 × 4 × 37 = 37 × 25 × 4 = 100 × 37 = 3700. But 25 + 4 × 37 = 25 + 148 = **173** — it has an addition, so it is different.

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**6** Ali makes tile patterns. Pattern n has n × n white tiles in a square, and 4 grey tiles at the corners. **[1]**



Pattern 1 Pattern 2 Pattern 3

How many tiles (white and grey) are in **Pattern 12**?

**Answer:** 148
**Notes / working:** Pattern 1: 1 + 4 = 5, Pattern 2: 4 + 4 = 8, Pattern 3: 9 + 4 = 13. Pattern 12: 12 × 12 + 4 = 144 + 4 = **148** tiles. Common mistake: 12 × 4 + 4 = 52 (the white tiles grow as square numbers, not by the same amount each time).

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**7** Here is part of a cinema timetable. **[1]**

| Film | Starts | Length |
|---|---|---|
| Desert Falcon | 14:35 | 1 hour 50 minutes |
| Space Cats | 15:10 | 1.4 hours |
| The Lost Pyramid | 16:45 | 2.25 hours |

(a) At what time does *Desert Falcon* end?

**Answer:** 16:25 or 1625 or 16.25 or 4:25pm or 4.25pm or 4:25 pm
**Notes / working:** 14:35 + 1 hour = 15:35. 15:35 + 50 minutes: 25 minutes takes you to 16:00, then 25 more minutes = **16:25**.

(b) *Space Cats* lasts 1.4 hours. Complete: 1.4 hours = 1 hour and ☐ minutes. **[1]**

> **Answer:** 24 minutes
> **Notes / working:** 0.4 of an hour = 0.4 × 60 minutes = **24 minutes**. So 1.4 hours = 1 hour 24 minutes. Common mistake: 40 minutes — an hour has 60 minutes, not 100.

(c) Hany watches *Desert Falcon*. After the film he walks to the bus stop, which takes 20 minutes. His bus leaves at 16:40. Will he catch the bus? Show how you know. **[1]**

> **Answer:** 1 mark: No, with a reason, e.g. 16:25 + 20 minutes = 16:45, which is after 16:40 / he is 5 minutes late. Follow through from their answer to (a).
> **Notes / working:** The film ends at 16:25. 16:25 + 20 minutes = 16:45. The bus leaves at 16:40, which is 5 minutes earlier, so **no**, Hany will miss the bus.

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**8** Mariam travels to school by bus. **[1]**

(a) A card for 12 bus journeys costs E£81.60. How much does one journey cost with the card?

> **Answer:** 6.8 E£
> **Notes / working:** 81.60 ÷ 12 = **E£6.80**. Check: 6.80 × 12 = 81.60 ✓.

(b) The bus travels at a steady speed. It travels 18 km in 24 minutes. How far does it travel in 40 minutes? Show your working. **[2]**

> **Answer:** 2 marks for 30 (km). If 0 marks: M1 for a correct method, e.g. 18 ÷ 24 (= 0.75 km per minute) then × 40; or 18 km in 24 min → 6 km in 8 min → 30 km in 40 min; or 40 ÷ 24 × 18.
> **Notes / working:** Find the distance in 1 minute: 18 ÷ 24 = 0.75 km. In 40 minutes: 0.75 × 40 = **30 km**. Or: 24 minutes ÷ 3 = 8 minutes for 6 km, and 40 minutes = 5 × 8 minutes, so 5 × 6 = 30 km. (M1 method, A1 30.)

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**9** A supermarket has a sale. **[1]**

(a) A 2 kg bag of rice usually costs E£64. In the sale, the price is reduced by 15%. What is the sale price?

> **Answer:** 54.4 E£
> **Notes / working:** 15% of 64 = 0.15 × 64 = 9.60. Sale price = 64 − 9.60 = **E£54.40**. (Or 85% of 64 = 0.85 × 64 = 54.40.) Common mistake: giving 9.60, the discount, instead of the new price.

(b) Pack A has 6 yogurts for E£37.20. Pack B has 8 yogurts for E£47.60. Which pack is better value? Show your working. **[2]**

> **Answer:** 2 marks: Pack B with both prices for one yogurt: A = E£6.20, B = E£5.95 (or another correct comparison, e.g. cost of 24 yogurts: A = E£148.80, B = E£142.80). M1 for one correct price per yogurt (6.20 or 5.95) or a correct method for both. Do not accept 'Pack B' with no working.
> **Notes / working:** Price of one yogurt: Pack A 37.20 ÷ 6 = E£6.20; Pack B 47.60 ÷ 8 = E£5.95. One yogurt is cheaper in **Pack B**, so Pack B is better value. (M1 for one correct unit price, A1 for both and Pack B.)

(c) A can of beans costs E£23.75. How much do 12 cans cost? **[1]**

> **Answer:** 285 E£
> **Notes / working:** 23.75 × 12 = **E£285**. Estimate: 24 × 12 = 288, so E£285 is sensible.

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**10** **[1]**

Which list is in order from **smallest to largest**?

- ☐ A) 0.69, 0.7, 72%, 3/4
- ☐ B) 0.7, 0.69, 72%, 3/4
- ☐ C) 0.69, 72%, 0.7, 3/4
- ☐ D) 3/4, 72%, 0.7, 0.69

> **Answer:** A) 0.69, 0.7, 72%, 3/4
> **Notes / working:** Change them all to the same form (hundredths): 0.69 = 69%, 0.7 = 70%, 72%, 3/4 = 75%. In order: 69%, 70%, 72%, 75%, so **0.69, 0.7, 72%, 3/4**. Common mistake: thinking 0.69 > 0.7 because 69 > 7.

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**11** Here is a recipe for koshari for **6 people**. **[1]**

| Ingredient | Amount |
|---|---|
| rice | 360 g |
| brown lentils | 240 g |
| macaroni | 300 g |
| tomato sauce | 450 ml |
| onions | 3 |

(a) How much rice is needed to make koshari for **10 people**?

> **Answer:** 600 g
> **Notes / working:** Rice for 1 person: 360 ÷ 6 = 60 g. For 10 people: 60 × 10 = **600 g**.

(b) Write the ratio **rice : lentils** in its simplest form. **[1]**

> **Answer:** 3:2
> **Notes / working:** rice : lentils = 360 : 240. Divide both by the highest common factor 120: **3 : 2**. Common mistake: 2 : 3 (wrong order — rice comes first).

(c) Karim has a 600 ml bottle of tomato sauce. He uses 3/4 of the bottle. How many millilitres does he use? **[1]**

> **Answer:** 450 ml
> **Notes / working:** 1/4 of 600 = 600 ÷ 4 = 150 ml, so 3/4 = 3 × 150 = **450 ml** — exactly the amount in the recipe.

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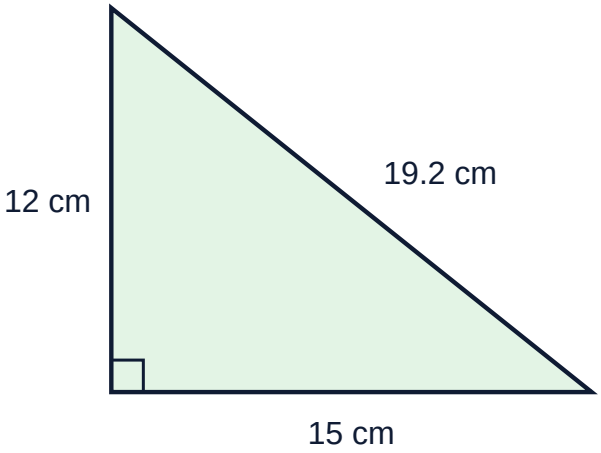
**12** Mona cooks koshari in a large pot. **[1]**

Which sentence is about the **capacity** of the pot?

- ☐ A) The metal of the pot takes up 600 $cm^3$ of space.
- ☐ B) The pot can hold 5 litres of food.
- ☐ C) The pot has a mass of 1.2 kg.
- ☐ D) The base of the pot has an area of 490 $cm^2$.

> **Answer:** B) The pot can hold 5 litres of food.
> **Notes / working: Capacity** is how much a container can hold, so 'the pot can hold 5 litres' is about capacity. The 600 $cm^3$ of metal is the **volume** of the pot itself (the space it takes up). Mass and area are different measures.

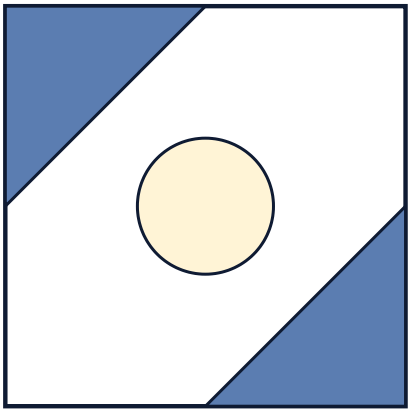
**13** Nour designs tiles for a kitchen wall. **[1]**




(a) One tile is a right-angled triangle. Work out its area.

**Answer:** 90 cm²
**Notes / working:** The two sides that meet at the right angle are 15 cm and 12 cm. A rectangle 15 cm by 12 cm has area 180 cm², and the triangle is half of it: 180 ÷ 2 = **90 cm²**. The 19.2 cm side is not used. Common mistake: 15 × 12 = 180 (forgetting to halve).

**[1]**



(b) Here is another of Nour's tiles. What is its order of rotational symmetry?

- ☐ A) order 1
- ☐ B) order 2
- ☐ C) order 3
- ☐ D) order 4

**Answer:** B) order 2
**Notes / working:** Turn the tile about its centre. After a half turn (180°) the shaded corner triangles swap places and the tile looks the same. After a quarter turn (90°) the shaded triangles are in the other two corners, so it looks different. It looks the same 2 times in a full turn: **order 2**.

(c) Nour covers **all** the faces of a cube-shaped box with square tiles. The edges of the box are 20 cm long. Each tile is 10 cm by 10 cm. How many tiles does she need? **[1]**

> **Answer:** 24
> **Notes / working:** Each face is a 20 cm × 20 cm square, so 2 × 2 = 4 tiles fit on one face. A cube has 6 faces: 6 × 4 = **24 tiles**. (Surface area = 6 × 400 = 2400 cm²; one tile is 100 cm²; 2400 ÷ 100 = 24.)

(d) Nour says, "Every square is also a rectangle." Is she correct? Explain your answer. **[1]**

> **Answer:** 1 mark: Yes, with a reason, e.g. a square has 4 right angles and opposite sides equal and parallel, so it has all the properties of a rectangle. Do not accept 'No, because a square has 4 equal sides' (this does not stop it being a rectangle).
> **Notes / working:** A rectangle is a quadrilateral with four right angles (so opposite sides are equal and parallel). A square has four right angles and opposite sides equal and parallel, so **yes**, every square is a rectangle — a special one with all four sides equal. (But not every rectangle is a square.)

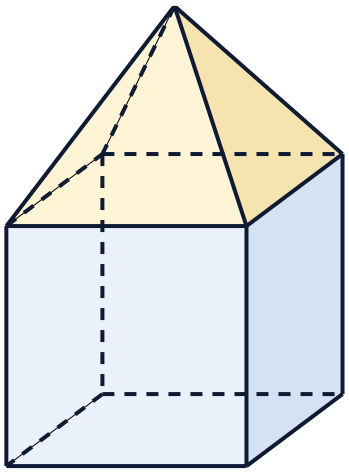
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**14** **[1]**

Sketch a net of a **triangular prism**. Draw it on paper.

> **Answer:** 1 mark: a net with 2 triangles and 3 rectangles that would fold to make a triangular prism, e.g. three rectangles in a row with one triangle on the top and one on the bottom (attached to the same or different rectangles), with matching edge lengths. Do not accept a net with the triangles on the same side so that they overlap when folded.
> **Notes / working:** A triangular prism has 5 faces: 2 triangles (the ends) and 3 rectangles. One correct net: draw 3 rectangles side by side in a row, then draw one triangle above the middle rectangle and one triangle below it. When you fold it, the rectangles wrap round and the triangles close the ends.

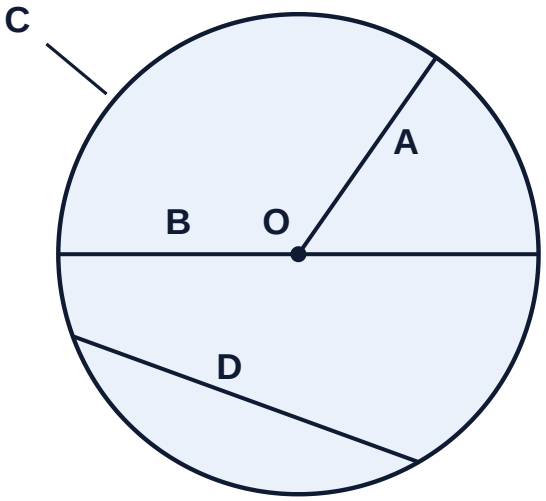
**15** This shape is made from a cube with a square-based pyramid on top. **[1]**



How many **faces** does the shape have?

**Answer:** 9
**Notes / working:** The cube has 6 faces, but its top face is covered by the pyramid, so 5 cube faces can be seen from outside. The pyramid's square base is joined to the cube, so it adds its 4 triangular faces. 5 + 4 = **9 faces**. Common mistake: 6 + 5 = 11 (counting the hidden faces where the two shapes join).

**16** Here is a circle with centre O. **[1]**




(a) Which letter shows a **radius** of the circle?

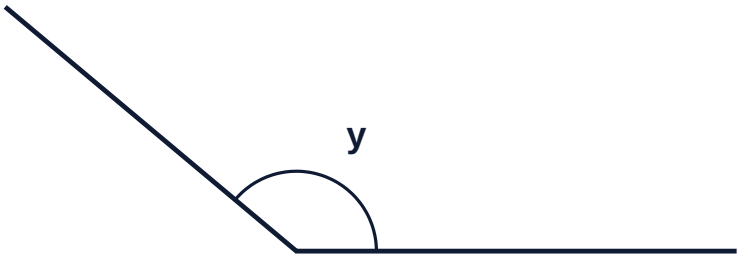
**Answer:** A
**Notes / working:** A radius goes from the centre to the circumference: that is line **A**. B goes right across through the centre (a diameter), C points to the circumference (the curved edge), and D joins two points on the circle but does not pass through the centre.

(b) Use compasses to draw a circle with a **diameter of 8 cm** on paper. Mark its centre. **[1]**

> **Answer:** 1 mark: a circle with radius 4 cm (diameter 8 cm), accurate to ± 2 mm, with the centre marked. Do not accept a circle of radius 8 cm.
> **Notes / working:** The diameter is 8 cm, so the radius is 8 ÷ 2 = **4 cm**. Open the compasses to 4 cm on a ruler, put the point on the centre and turn them all the way round. Check: the line across through the centre measures 8 cm.

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**17** **[1]**

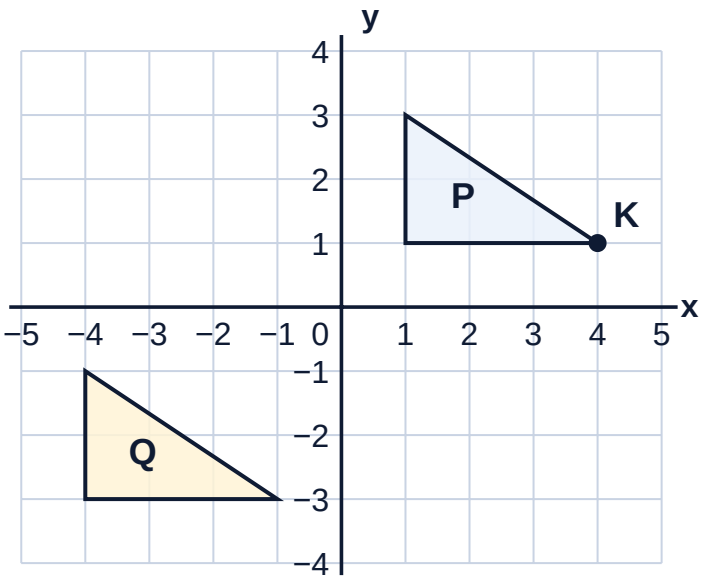



Which is the best **estimate** of the size of angle **y**?

- ☐ A) 40°
- ☐ B) 90°
- ☐ C) 140°
- ☐ D) 220°

> **Answer:** C) 140°
> **Notes / working:** The angle is bigger than a right angle (90°) but smaller than a straight line (180°), so it is obtuse. It is a little more than halfway between 90° and 180°: **140°**. 40° is acute and 220° is reflex.

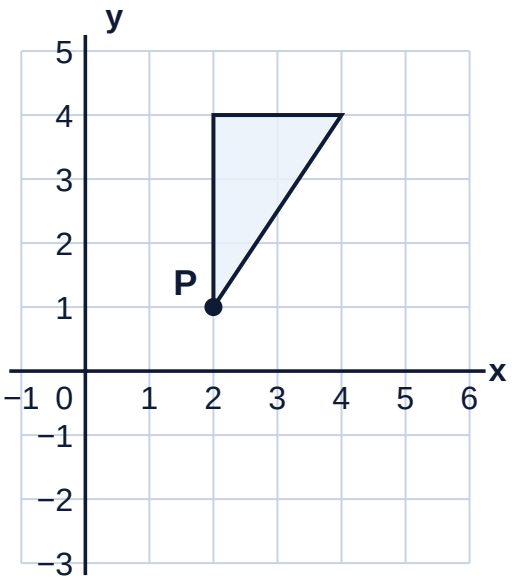
**18** Triangle P is translated to triangle Q. **[1]**




Point K on triangle P moves to a corner of triangle Q. Write the coordinates of this corner.

**Answer:** (-1,-3) or (−1,−3) or -1,-3 or −1,−3 or (-1, -3) or (−1, −3)
**Notes / working:** The translation moves every point 5 squares left and 4 squares down (for example, (1, 1) moves to (−4, −3)). K is at (4, 1): 4 − 5 = −1 and 1 − 4 = −3, so K moves to **(−1, −3)**.

**19** Here is a triangle on a coordinate grid. **[1]**




Copy the grid and rotate the triangle **90° clockwise** about the vertex **P**. Draw the new triangle.

**Answer:** 1 mark: image triangle with vertices (2, 1), (5, 1) and (5, −1). Do not accept a 90° anticlockwise rotation (vertices (2, 1), (−1, 1), (−1, 3)).
**Notes / working:** P stays where it is: (2, 1). The vertical side from P goes 3 up to (2, 4); after a quarter turn clockwise it goes 3 to the right, to (5, 1). The corner (4, 4) is 2 right and 3 up from P; after the turn it is 3 right and 2 down from P, at (5, −1). The new triangle has vertices **(2, 1), (5, 1) and (5, −1)**. Tracing paper helps: trace, hold the point at P, turn a quarter turn clockwise.

**20** Ten children wrote down how many pets they have at home. **[1]**
2, 0, 1, 16, 1, 3, 0, 4, 1, 2
(Omar has 16 fish in a fish tank.)

(a) Work out the **mean** number of pets.

**Answer:** 3
**Notes / working:** Total = 2 + 0 + 1 + 16 + 1 + 3 + 0 + 4 + 1 + 2 = 30. Mean = 30 ÷ 10 = **3** pets.

(b) Work out the **median** number of pets. **[1]**

**Answer:** 1.5
**Notes / working:** In order: 0, 0, 1, 1, **1, 2**, 2, 3, 4, 16. There are 10 values, so the median is halfway between the 5th and 6th values: (1 + 2) ÷ 2 = **1.5**. Common mistake: finding the middle of the unordered list.

(c) Which average, the mean or the median, better describes the number of pets a typical child has? Give a reason. **[1]**

> **Answer:** 1 mark: the median, with a reason, e.g. the mean is made bigger by Omar's 16 fish / 16 is much larger than the other values / only 2 of the 10 children have more than 3 pets. Do not accept 'median' with no reason.
> **Notes / working:** The **median** (1.5) is better. Omar's 16 fish make the total large, so the mean (3) is higher than what most children have — 8 of the 10 children have fewer than 3 pets. The median is not affected by one very large value.

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**21** Class 6B plans an investigation about pets. **[1]**

Which of these is **continuous** data?

- ☐ A) the mass of each pet in kilograms
- ☐ B) the number of pets each child has
- ☐ C) the type of each pet
- ☐ D) the number of legs each pet has

> **Answer:** A) the mass of each pet in kilograms
> **Notes / working:** Continuous data is **measured** and can take any value in a range (2.35 kg, 2.4 kg …), so **mass** is continuous. The number of pets and the number of legs are counted (discrete data); the type of pet is a word (categorical data).

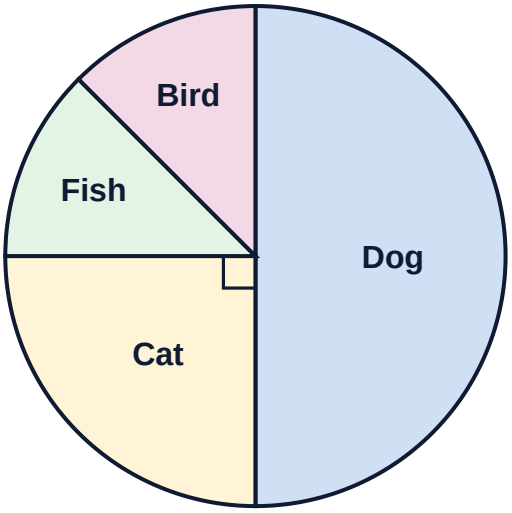
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**22** Salma weighs her puppy every week for 8 weeks to see how its mass changes. **[1]**

Which would be the best way to show her results: a **Venn diagram**, a **bar chart** or a **line graph**? Explain your choice.

> **Answer:** 1 mark: line graph, with a reason, e.g. mass is continuous data that changes over time / it shows how the mass goes up week by week / you can see the trend. Do not accept 'line graph' with no reason.
> **Notes / working:** A **line graph** is best. Mass is continuous data measured over time, and joining the points shows how the puppy's mass changes from week to week (the trend). A Venn diagram sorts things into groups, and a bar chart compares separate categories.

**23** 40 children were asked about their favourite pet. The pie chart shows the results. **[1]**




How many children chose **cat**?

**Answer:** 10
**Notes / working:** The cat sector is a right angle, which is a quarter of the circle (90° out of 360°). A quarter of 40 = 40 ÷ 4 = **10** children. (Dog is half: 20 children; fish and bird are an eighth each: 5 children each.)

**24** Yara rolls an ordinary dice numbered 1 to 6 once. **[1]**

Which two events are **mutually exclusive** (cannot happen at the same time)?

- ☐ A) getting an even number and getting a 4
- ☐ B) getting a number less than 3 and getting an odd number
- ☐ C) getting a 6 and getting an odd number
- ☐ D) getting a number greater than 2 and getting an even number

**Answer:** C) getting a 6 and getting an odd number
**Notes / working:** 6 is even, so you cannot get a 6 **and** an odd number on one roll: these events are **mutually exclusive**. The others can happen together: a 4 is even; a 1 is less than 3 and odd; a 4 or 6 is greater than 2 and even.

**25** Nadia spins a spinner 20 times and gets red 13 times. Karim spins the same spinner 500 times and gets red 260 times. **[1]**

Whose results give a better estimate of the probability of getting red? Explain why.

**Answer:** 1 mark: Karim, with a reason, e.g. he did many more spins / more trials give a more reliable estimate. Do not accept 'Karim, because he got more reds'.
**Notes / working: Karim's** results are better. With only 20 spins, chance can change the results a lot. With a **large number of trials** (500), the proportion of reds (260 out of 500, about half) gets closer to the true probability.